

NIKHIL YENGALA REDDY

BIOINFORMATICS | GENOMIC DATA ANALYSIS

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PROFESSIONAL SUMMARY

Bioinformatician with an MS in Bioinformatics from Johns Hopkins University with practical knowledge of human whole-genome sequencing data, FASTQ file format, BWA, SAMtools, and HPC systems. Assist with public health genomics analysis and custom pipeline development and conduct regional bioinformatics training. Technical skills are Python, Java, Bash, R, and Linux/Unix.

TECHNICAL SKILLS

Programming and Data: Python, Pandas, R, SQL, Bash, Jupyter, ggplot2, ggtree

Sequencing and Analysis: Human WGS, FASTQ, BWA, SAMtools, NCBI, Biopython, Bioconductor

Computing Environments: High-performance computing (HPC), Linux/UNIX, Conda, Docker, GitHub

PROFFESIONAL EXPERIENCE

Florida Department of Health | AMD Bioinformatician

2024 - Present

Bureau of Public Health Laboratories | Advanced Molecular Detection | Remote

- Promote WGS studies and carry out analyses on NCBI and lab data sets for public health research.
- Provide remote training to public health laboratories in the Southeast region about workflow processes like Sanibel, Talbot PHoeNix, Daytona, Sarek_Mic, MIRA-NF and many others.
- Help develop personalized bioinformatics pipeline to perform analysis in public health laboratories.
- Train the lab members to use Bash for sequence data analysis with SRA Human Scrubber and SeqSender, report writing, and phylogenetic tree construction with R and ggplot2 and ggtree packages.
- Created the BPHL Molecular Bioinformatics website to centralize pipeline resources and regional training materials.

Thermo Fisher Scientific | Associate Scientist

2024

Madison, WI

- Performed sample preparation and analysis, maintained electronic laboratory notebooks, and completed documentation in accordance with SOPs and regulatory guidelines.

RESEARCH EXPERIENCE

San Jose State University | Bioinformatics Research

Department of Microbiology and Bioinformatics

- Investigated multidrug resistance genes in the major facilitator superfamily using sequence alignment and phylogenetic analysis to examine evolutionary relationships.
- Used Biopython and UniProt to study related sequences and PyMOL to examine MDR protein structures.

RESEARCH EXPERIENCE

MS, Bioinformatics | Johns Hopkins University | 2023

BS, Molecular Biology; Minor, Chemistry | San Jose State University | 2021